Target : Learning Brownian Motion /Stochastic Differential Equation

(Learning level: Medium to Hard)

- *Brownian Motion*
  - ❖ Einstein's solution
- *Langevin's Approach*

  (Learning level: medium)

- *Numerical Techniques*

  (Learning level: Medium)

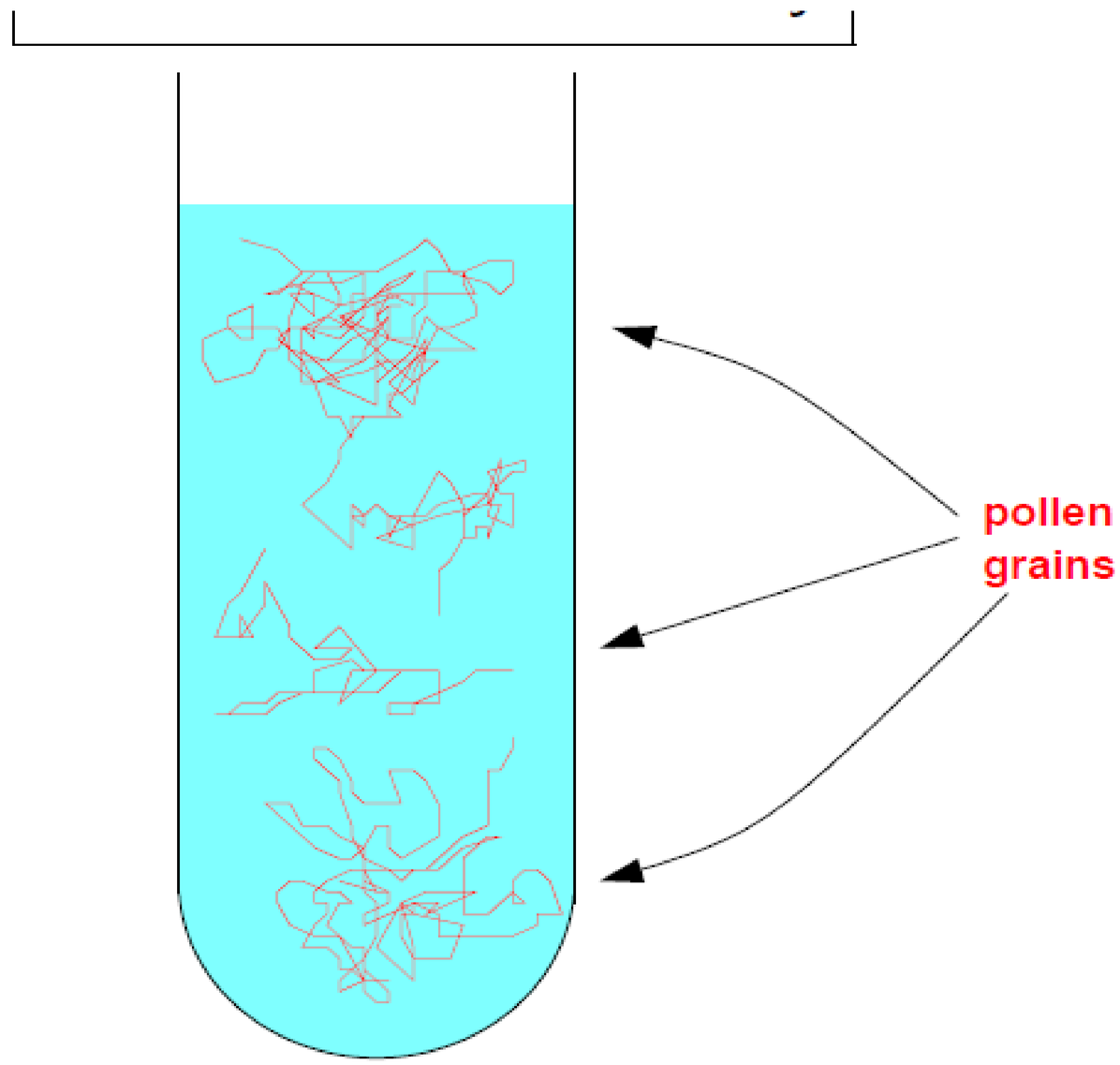
C.W. Gardiner

Handbook of Stochastic Methods

for Physics, Chemistry and the Natural Sciences

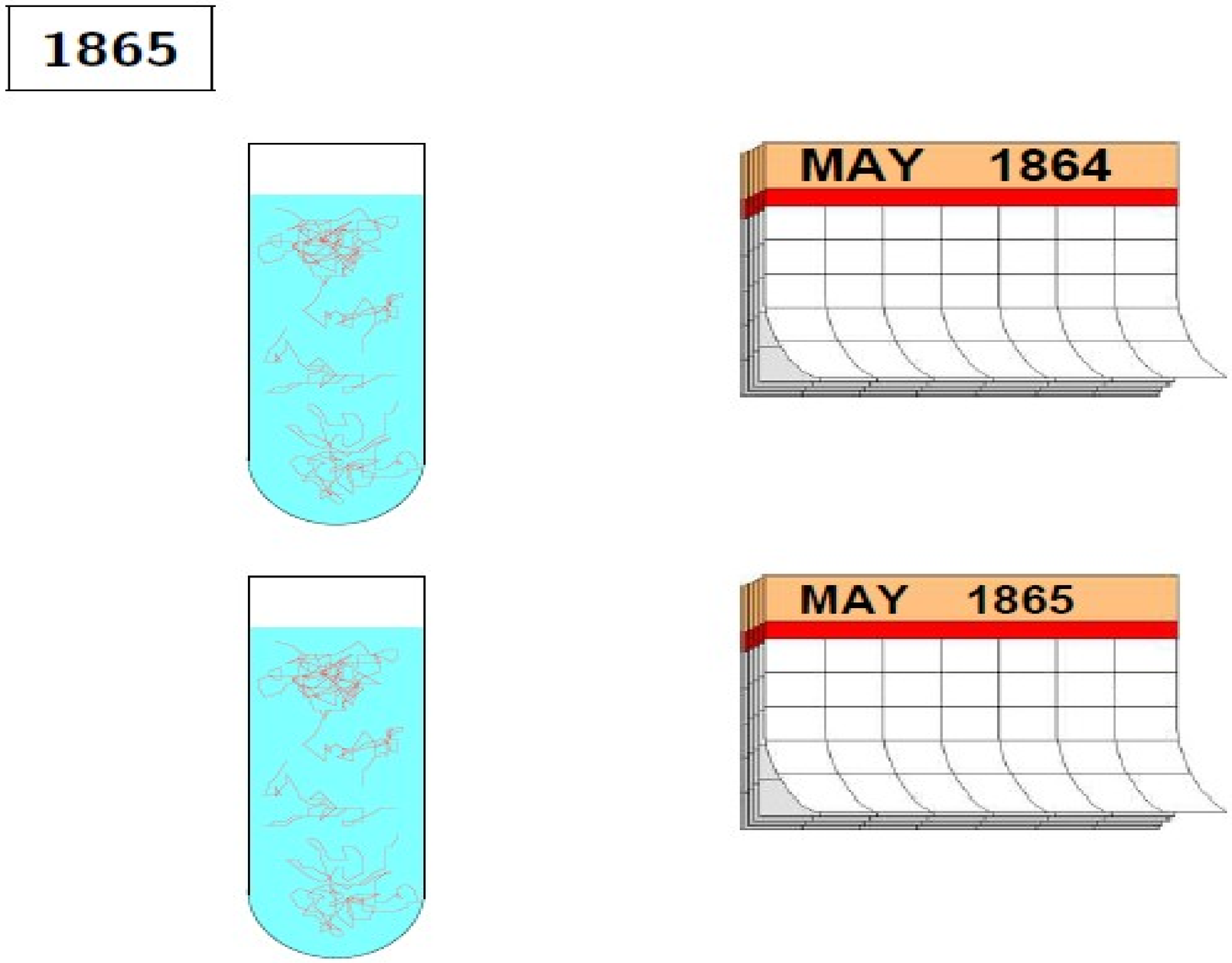
Simo Särkkä

Brownian Motion



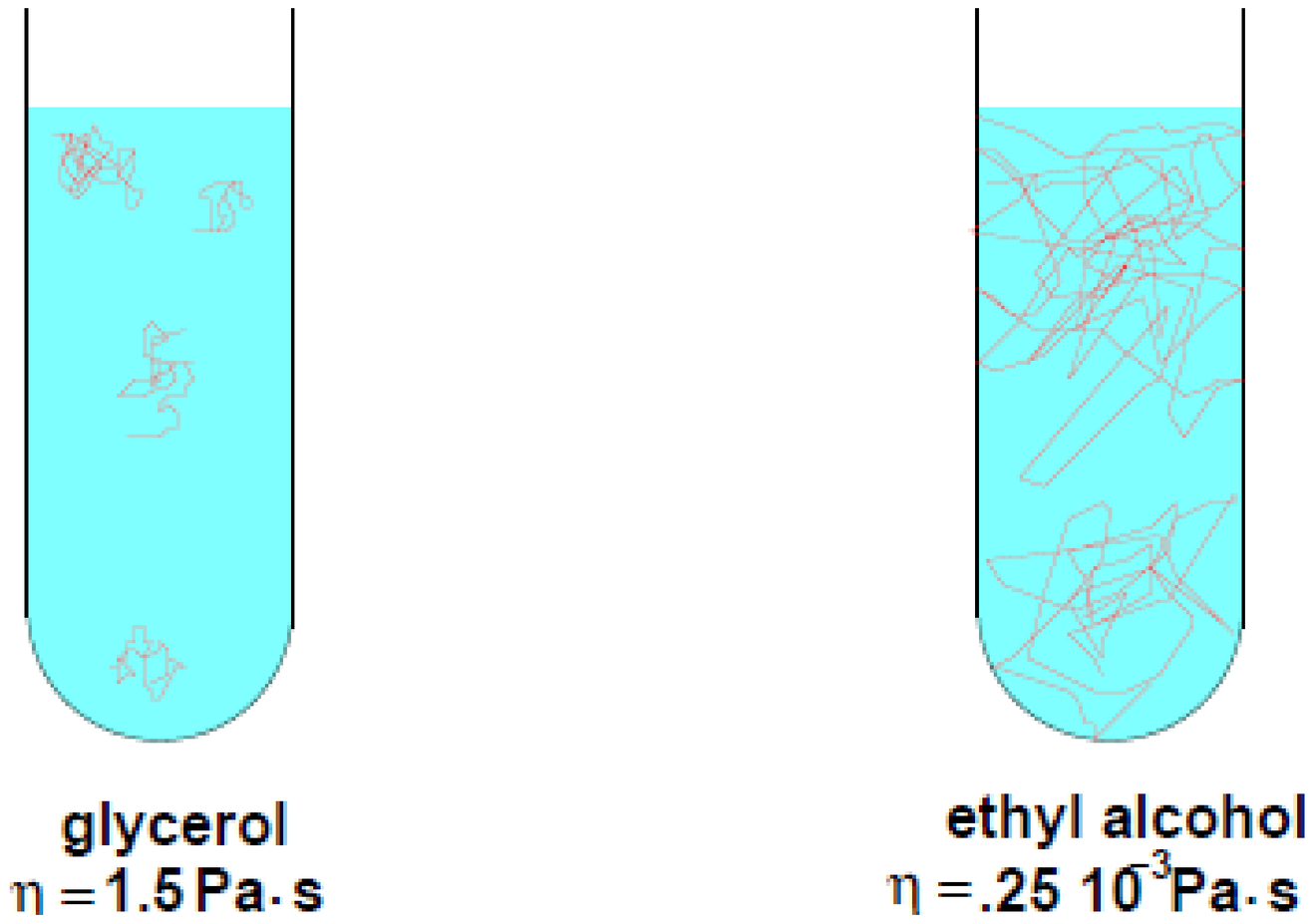
In 1827 Robert Brown, a Scottish botanist and curator of the British Museum, observed that pollen grains suspended in water, instead of remaining stationary or falling downwards, would trace out a random zig-zagging pattern. This process, which could be observed easily with a microscope, gained the name of **Brownian motion.**

Brownian Motion



It is discovered that no matter how long the particles are left in the liquid in a sealed container, they never come to rest or even slow down. The experimenters observed a system of pollen motes in a sealed volume of liquid over a period of one year, and Brown's zig-zagging effect continued undiminished.

Brownian Motion



It is discovered that the speed with which the pollen motes disperse was slower in more viscous liquid and faster in less viscous liquid.

Brownian Motion
Einstein's explanation

Thermal Equilibrium

The most **basic axiom of thermodynamics** states that **two systems** interacting with one another will eventually approach the **same temperature** after a sufficiently **long time** has passed. Boltzmann showed that this principle could be derived from **statistical mechanics** under broad conditions.

Principle of Equipartition

This principle refers to a gas in which the molecules have **internal degrees of freedom** – internal structures which can move, oscillate, or vibrate according to **mechanical forces** within the molecule.

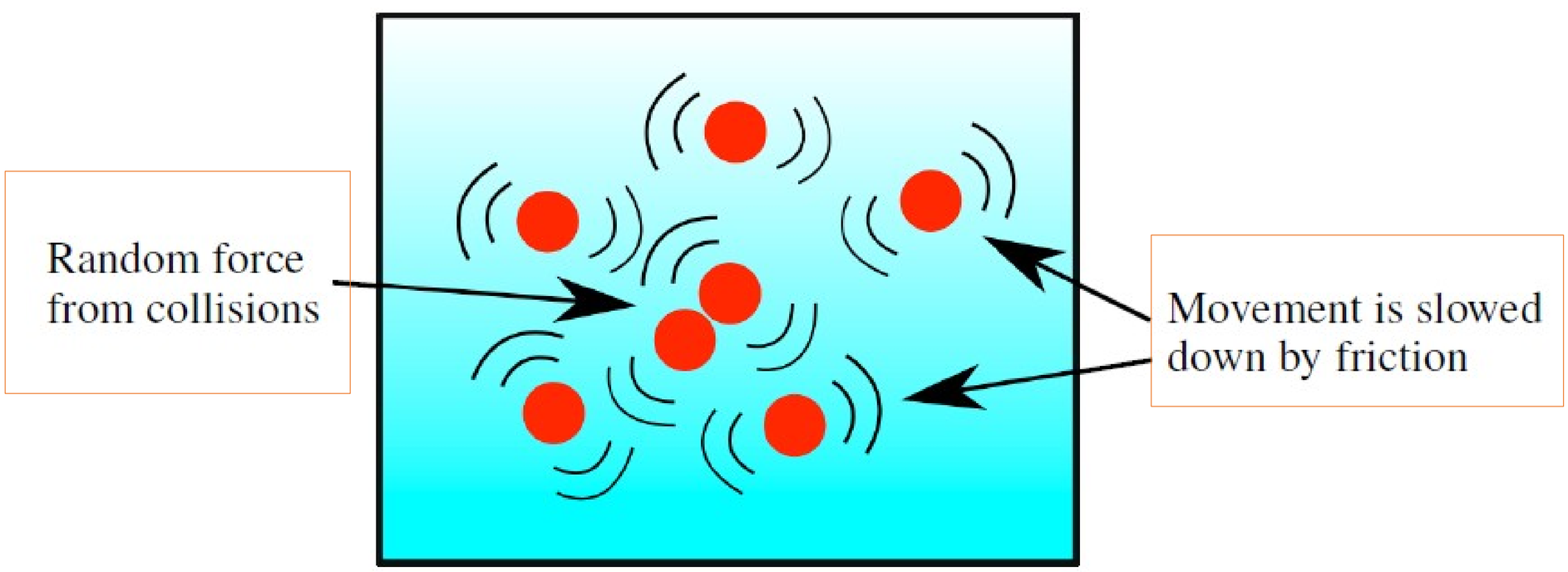
The principle of equipartition states that for a gas in thermal equilibrium, the average energy stored in **each available degree of freedom** will be equal to the **overall kinetic energy** of the molecule's **center of mass**.

With these two ideas, we can understand Einstein's theory of Brownian motion.

Langevin's model of Brownian motion

A couple of years after Einstein's contribution Langevin (1908) presented an alternative construction of Brownian motion which leads to the same macroscopic properties. The reasoning in the article, which is outlined in the following, was more mechanical than in Einstein's derivation.

- Consider a small particle suspended in liquid.
- Assume that there are two kinds of forces acting on the particle:



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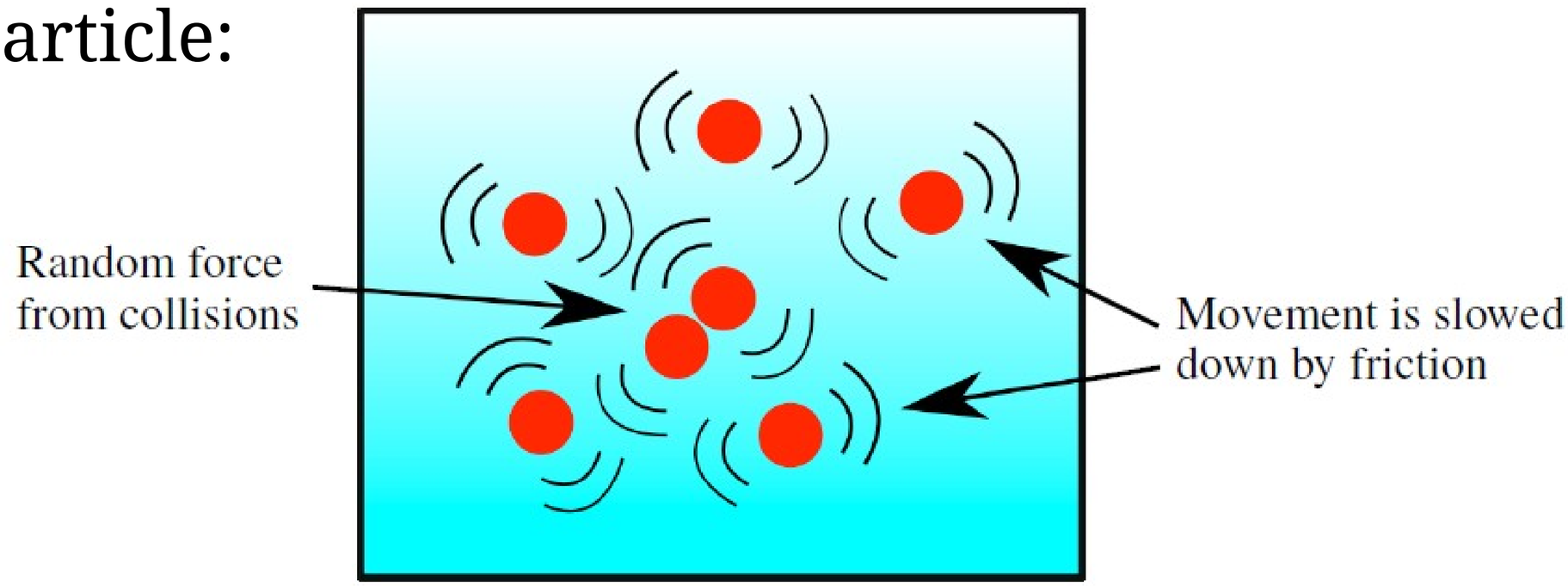

- Mass of the particle m .
- **A viscous drag:** Coming from the macroscopic hydrodynamics, this is $-6\,\pi\eta a\frac{dx}{dt}$; where η being the viscosity and a is the diameter of the particle. We assume the particle as spherical.

- **Fluctuating force:** X represents random collisions: ceaseless impacts of liquid: impacts on the Brownian particle.

(This force should be positive or negative with equal probability.)

- Thus, the equation of motion of the particle can be captured by

$$m\frac{d^2x}{dt^2} = -6\pi\eta a\frac{dx}{dt} + X$$

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Recall that, $\frac{1}{2}\frac{d(x^2)}{dt} = \frac{dx}{dt}\,x$

And, $\frac{1}{2}\frac{d^2(x^2)}{dt^2} = \frac{d}{dt}\left(\frac{1}{2}\frac{d(x^2)}{dt}\right) = \frac{d}{dt}\left(\frac{dx}{dt}x\right) = \frac{d^2x}{dt^2}x + \left(\frac{dx}{dt}\right)^2$

Thus, $\frac{1}{2}\frac{d^2(x^2)}{dt^2} - \left(\frac{dx}{dt}\right)^2 = \frac{d^2x}{dt^2}x$

- Thus, if we multiply the equation of motion with x, we obtain

$$mx\frac{d^2x}{dt^2} = -6\pi\eta x\frac{dx}{dt} + Xx$$

- Substitute the above identities, we get

$$\frac{m}{2}\frac{d^2(x^2)}{dt^2} - mv^2 = -3\pi\eta a\frac{d(x^2)}{dt} + Xx$$

Langevin's model of Brownian motion

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- We now average over a large number of different particles, and we use the following theorem from statistical mechanics:

The mean kinetic energy of the Brownian particles should follow $\left\langle \frac{1}{2}mv^2 \right\rangle = \frac{1}{2}k_B T$ (in equilibrium. T is abs. Temp. and k is the Boltzmann's constant). Thus

$$\frac{m}{2}\frac{d^2\langle x^2\rangle}{dt^2} + 3\pi\eta a\frac{d\langle x^2\rangle}{dt} = k_B T + \langle Xx\rangle$$

"....of irregularity of the quantity X" we can set $\langle Xx\rangle$ equal to zero

- One can find the general solution as

$$\frac{d\langle x^2\rangle}{dt} = \frac{k_B T}{3\pi\eta a} + C\exp\left(-\frac{6\pi\eta a t}{m}\right)$$

Langevin's model of Brownian motion

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C is an arbitrary constant. Langevin estimated that the decaying term vanishes quickly ($10^{-8}s$), "which for any practical observation at that was essentially immediately". Thus we can neglect this term and integrate once more to get

$$\langle x^2\rangle = \frac{k_B T}{3\pi\eta a} t$$

$$\text{and } D = \frac{k_B T}{6\pi\eta a}$$

a result which Einstein derived in the same paper but by independent means.

Langevin's equation was the first example of the *stochastic differential equation*—a differential equation with a random term X and hence whose solution is, in some sense, a random function. Each solution of Langevin's equation represents a different random trajectory and, using only rather simple properties of X (his fluctuating force), measurable results can be derived.

Langevin's model of Brownian motion

$$\langle x^2 \rangle = \frac{k_B T}{3\pi\eta a} t$$

- The motion increases with temperature T.
- It decreases with increasing viscosity η.
- It also decreases with the size of the particles.

- In agreement with the observed behaviour of Brownian motion.

Euler-Maruyama Method and Brownian motion (Wiener process)

Euler-Maruyama method comes from the properties of Brownian motion (Wiener process).

A "Brownian motion" or "Wiener process" refers to a mathematical model describing a continuous, random movement of a particle, where the particle's displacement over any time interval follows a normal distribution with a mean of zero and variance proportional to the time interval; essentially, it's a random walk with infinitely small steps, often visualized as a jagged line on a graph, and named after the botanist Robert Brown who first observed this phenomenon in pollen grains suspended in water.

Euler-Maruyama Method and Brownian motion (Wiener process)

It is used to approximate solution of SDEs of the form:

$$dX_t = f(X_t, t)dt + g\,(X_t, t)dW_t$$

- X_t is the state variable.
- $f(X_t, t)$ is the drift term (deterministic part)
- $g\,(X_t, t)$ is the diffusion term (random part)
- W_t is Wiener process (a fancy way of saying Brownian motion)

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For small time steps Δt, Euler-Maruyama update rule is:

$$X_{t+\Delta t} = X_t + f(X_t, t)\Delta_t + gf(X_t, t)\Delta W_t,$$

Where ΔW_t is normally distributed random variable with mean 0 and variance Δt

Why?

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Euler-Maruyama Method

Why $\Delta W_t \sim N(0, \Delta t)$?

A Wiener process W_t has the following properties

- $W_0 = 0$ starts at zero.
- **Independent increments.** The change $W_{t+\Delta t} - W_t$ are independent non-overlapping time intervals.
- **Normally distributed increments:** $W_{t+\Delta t} - W_t \sim N(0, \Delta t)$

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Key Idea:

- If we approximate dW_t, over a small step Δt, we need to match statistical behaviour.
- Since, Werner process increments scale as $N(0, \Delta t)$, it follows that

$$\Delta W_t = \sqrt{\Delta t}\, Z$$

Where $Z \sim N(0, 1)$ is a standard normal random variable.

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Why $\sqrt{\Delta t}$ specifically?

- If $Z \sim N(0, 1)$, then multiplying by $\sqrt{\Delta t}$ gives $N(0, \Delta t)$, which matches the properties of ΔW_t.

- Intuitively, this mean Brownian motion scales with the square root of time rather that linearly.

Euler-Maruyama Method: Pseudo code

```
Initialize parameters:
T = total simulation time
N = number of steps
dt = T / N (time step)
X[0] = initial value
mu = drift coefficient
sigma = diffusion coefficient

Generate Wiener process:
for i = 1 to N:
Z[i] = random normal sample from N(0,1)
dW[i] = sqrt(dt) * Z[i]
W[i] = W[i-1] + dW[i] (Cumulative sum)

Apply Euler-Maruyama method:
for i = 1 to N:
X[i+1] = X[i] + mu * X[i] * dt + sigma * X[i] * dW[i]

Plot X vs time
```